MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)
SEM 2 2025/2026

LAB 09:
BLUETOOTH DATA INTERFACING

SECTION 1
GROUP 10

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Abstract

This experiment was conducted to develop a wireless monitoring and control system using Bluetooth communication with the ESP32 microcontroller for Internet of Things (IoT) applications. The system combines a sensor and a servo motor to allow wireless data transfer and remote control through a smartphone application. The ESP32 is used to read the sensor data and send the information to the Bluetooth terminal application on the smartphone in real time. At the same time, commands sent from the smartphone are received by the ESP32 to control the movement of the servo motor. Through this experiment, Bluetooth communication using the Serial Bluetooth feature of the ESP32 was successfully implemented for both monitoring and control purposes. The results obtained show that the system was able to transmit sensor readings accurately and respond properly to user commands. Overall, the experiment proves that Bluetooth technology can be effectively used in simple IoT-based monitoring and automation systems.

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Introduction

The purpose of this experiment is to develop a wireless monitoring and control system using Bluetooth communication between an ESP32 microcontroller and a smartphone. In modern Internet of Things (IoT) applications, Bluetooth technology is widely used for short-range wireless communication because it offers low power consumption, fast data transmission, and easy connectivity with mobile devices.

In this experiment, the ESP32 board is used because it has built-in Bluetooth capability, allowing it to send sensor data wirelessly and receive commands from a smartphone application. A potentiometer sensor is connected to the ESP32 to provide analog input data, while an LED or servo motor can be controlled remotely through Bluetooth commands. The communication is achieved using Bluetooth Serial communication, where the smartphone acts as a wireless terminal to monitor sensor readings and send control instructions.

This experiment helps students understand the integration of sensors, actuators, and wireless communication in IoT systems. It also demonstrates how Bluetooth technology can be implemented for remote monitoring and automation applications such as smart homes, industrial monitoring, and robotic systems.

Material and Equipment

The materials and equipment used in this experiment are:

1. Espressif Systems ESP32 Development Board
2. Potentiometer sensor
3. Servo motor / LED
4. Smartphone (Android or iOS)
5. Breadboard
6. Jumper wires
7. USB cable for programming and power supply
8. Arduino IDE software
9. Bluetooth Serial Terminal application

Experimental Setup

An ESP32 connected to a potentiometer sensor and a servo motor/LED on a breadboard make up the experiment setup. The Arduino IDE is used to program the ESP32, which is powered by a USB connection. The ESP32's integrated Bluetooth module enables Bluetooth communication.

To provide adjustable sensor readings, the potentiometer is attached to the ESP32's analog input pin. For remote control operation, one of the digital output pins is connected to the servo motor or LED. The ESP32 gadget is connected with a smartphone that has a Bluetooth terminal app.

Throughout operation, the ESP32 continuously reads the sensor value and sends the information via Bluetooth Serial to the smartphone. Additionally, the user can control the servo motor position or turn the LED on and off by sending commands from the smartphone application. The entire configuration shows a basic Bluetooth-based wireless IoT monitoring and control solution.

Methodology

This experiment was carried out to develop a Bluetooth-based wireless monitoring and control system using the ESP32 microcontroller. First, the ESP32 was connected to a servo motor and a potentiometer on a breadboard using jumper wires. The potentiometer was used as the input sensor to provide analog data, while the servo motor acted as the output actuator. The servo signal pin was connected to GPIO13 and the potentiometer signal pin was connected to GPIO34. All power and ground connections were properly shared with the ESP32 to ensure stable operation.

Next, the Arduino IDE software was used to write and upload the Bluetooth communication program into the ESP32. The program was designed to continuously read the potentiometer value and transmit the sensor data to a smartphone through Bluetooth communication. At the same time, the ESP32 was programmed to receive commands from the smartphone to control the rotation angle of the servo motor.

After the code was uploaded successfully, the Bluetooth feature on the ESP32 was activated and paired with a smartphone using the Serial Bluetooth Terminal application. Once the connection was established, the smartphone terminal displayed the potentiometer readings in real time. Different commands were then sent from the smartphone to control the movement of the servo motor wirelessly.

Finally, observations were recorded based on the responsiveness of the Bluetooth communication, the accuracy of the sensor readings displayed on the smartphone, and the ability of the servo motor to respond correctly to the commands received from the Bluetooth terminal.

Data collection

In this experiment, we conducted two tests, the potentiometer input test and Serial Bluetooth command override test.

Potentiometer value	Expected angle	Angle shown in serial display
0	0	0
1151	51	50
2254	99	99
4095	180	180

Potentiometer Input Test Data

Potentiometer value	Potentiometer angle	Bluetooth command	Expected angle	Angle shown in serial display
0	0	L	0	0
0	0	M	90	90
0	0	R	180	180

Serial Bluetooth Command Override Test Data

```
17:41:53.267 M
17:41:53.923 Pot: 0 | Angle: 90
17:41:54.943 Pot: 0 | Angle: 0
17:41:55.941 Pot: 0 | Angle: 0
17:41:56.919 Pot: 0 | Angle: 0
17:41:57.922 Pot: 0 | Angle: 0
17:41:59.324 Pot: 0 | Angle: 0
17:41:59.926 Pot: 0 | Angle: 0
17:42:00.924 Pot: 0 | Angle: 0
17:42:01.920 Pot: 0 | Angle: 0
17:42:02.938 Pot: 0 | Angle: 0
17:42:03.922 Pot: 0 | Angle: 0
17:42:04.942 L
17:42:05.001 Pot: 0 | Angle: 0
17:42:05.923 Pot: 0 | Angle: 0
17:42:06.924 Pot: 0 | Angle: 0
17:42:07.935 Pot: 0 | Angle: 0
17:42:08.965 Pot: 0 | Angle: 0
17:42:09.923 Pot: 0 | Angle: 0
17:42:11.204 Pot: 0 | Angle: 0
17:42:11.922 Pot: 0 | Angle: 0
17:42:12.352 R
17:42:12.955 Pot: 0 | Angle: 180
17:42:13.937 Pot: 0 | Angle: 0
17:42:14.942 Pot: 0 | Angle: 0
17:42:15.956 Pot: 0 | Angle: 0
```

Serial Bluetooth Display for Bluetooth Command Override Test

```
17:43:10.923 Pot: 0 | Angle: 0
17:43:11.943 Pot: 247 | Angle: 10
17:43:13.081 Pot: 658 | Angle: 28
17:43:14.069 Pot: 1151 | Angle: 50
17:43:16.049 Pot: 1424 | Angle: 62
17:43:16.051 Pot: 1743 | Angle: 76
17:43:16.932 Pot: 2254 | Angle: 99
17:43:17.981 Pot: 2670 | Angle: 117
17:43:18.953 Pot: 3184 | Angle: 139
17:43:19.933 Pot: 3695 | Angle: 162
17:43:20.926 Pot: 4095 | Angle: 180
17:43:21.990 Pot: 4095 | Angle: 180
17:43:23.476 Pot: 4095 | Angle: 180
17:43:23.934 Pot: 4095 | Angle: 180
17:43:24.930 Pot: 4095 | Angle: 180
17:43:25.960 Pot: 4095 | Angle: 180
17:43:26.971 Pot: 4095 | Angle: 180
17:43:27.976 Pot: 4095 | Angle: 180
17:43:29.414 Pot: 4095 | Angle: 180
17:43:30.407 Pot: 3901 | Angle: 171
17:43:30.939 Pot: 3260 | Angle: 143
17:43:31.950 Pot: 2723 | Angle: 119
17:43:32.953 Pot: 2484 | Angle: 109
17:43:33.942 Pot: 2124 | Angle: 93
17:43:34.941 Pot: 1507 | Angle: 66
17:43:36.345 Pot: 1084 | Angle: 47
17:43:37.335 Pot: 951 | Angle: 41
17:43:37.940 Pot: 481 | Angle: 21
17:43:38.940 Pot: 0 | Angle: 0
```

Serial Bluetooth Display for Potentiometer Test

Data analysis

The collected data from potentiometer test shows a direct proportional relationship between the potentiometer reading and the servo motor angle. When the potentiometer value increases, the angle of servo will increase as the ESP32 reads the analog values of potentiometer from 0 to 4095 and maps them to 0° to 180°. As for the bluetooth command test, the servo moves based on the command given where L, M and R force the servo to move to the angle 0°, 90° and 180° accordingly. Then, the servo will automatically move back according to the potentiometer value as potentiometer controls the servo continuously. This demonstrates a dual mode control where the potentiometer controls the servo automatically based on its value while the servo can be controlled manually via smartphone by giving command to directly control the servo position. Based on the data, this experiment has accurate data transmission as the output data in serial display align with the expected data. This shows that the bluetooth communication was stable and it showed a real time update every second.

Results

The experiment was successfully combining automation and manual remote control by integrating the sensor and communication by bluetooth to control the output. The potentiometer sensor was able to provide analog input values ranging from 0 to 4095, which were accurately converted into servo motor angles between 0° and 180° using the programmed mapping function. As the potentiometer value increased, the servo angle also increased proportionally, showing correct sensor-to-actuator response. Bluetooth communication between the ESP32 and smartphone was established successfully, allowing real-time transmission of potentiometer values and servo angle data to the smartphone terminal every second. The system also responded correctly to Bluetooth control commands, where 'L' moved the servo to 0°, 'M' moved it to 90°, and 'R' moved it to 180°, regardless of the potentiometer position. This confirmed that Bluetooth commands could override sensor-based control effectively. Overall, the system demonstrated stable wireless communication, accurate sensor monitoring, and reliable servo motor control where it successfully integrated sensor input, Bluetooth data transmission, and remote actuator control into a functional IoT monitoring and control system.

Discussion

This experiment successfully demonstrated the integration of Bluetooth communication into an IoT application by combining a potentiometer sensor, ESP32 microcontroller, smartphone, and servo motor into one wireless monitoring and control system. The potentiometer acted as the sensor input by providing analog values between 0 and 4095, then were processed by the ESP32 and converted into servo motor angles ranging from 0° to 180°. The results showed that as the potentiometer value increased, the servo angle also increased

proportionally, confirming that the mapping function and sensor-to-actuator relationship operated accurately.

The real-time sensor readings and servo angle data was transmitted to the smartphone terminal every second via communication between bluetooth and ESP32, enabling remote monitoring of system performance. Bluetooth was also used as a wireless control medium, where commands sent from the smartphone such as ‘L’, ‘M’, and ‘R’ successfully controlled the servo motor by moving it to predefined angles of 0°, 90°, and 180°. This demonstrated that Bluetooth can support both monitoring and control functions in IoT systems.

Through this experiment, Bluetooth was incorporated into IoT applications by serving as a short-range wireless communication technology that connects smart devices, sensors, and actuators for real-time data exchange. In practical IoT applications, Bluetooth can be used in smart homes, healthcare devices, industrial automation, and remote-controlled systems where nearby wireless communication is required. For example, Bluetooth can allow users to monitor temperature sensors, control home appliances, or operate robotic systems directly from smartphones without physical connections.

In conclusion, this experiment proved that Bluetooth can be effectively incorporated into IoT applications by enabling wireless communication between sensors, controllers, and user devices. The successful integration of sensing, data transmission, remote monitoring, and actuator control demonstrated how Bluetooth supports the development of simple, efficient, and practical IoT systems.

Conclusion

In conclusion, the experiment successfully demonstrated the implementation of Bluetooth communication between an ESP32 and a smartphone for wireless monitoring and control applications. The ESP32 was able to transfer sensor readings to the smartphone and receive control commands through Bluetooth Serial communication.

The experiment also showed how Bluetooth technology can be integrated into IoT systems for real-time data monitoring and remote device control. By combining sensors, actuators, and wireless communication, the system provides a simple and effective solution for automation applications.

Overall, the experiment's goals were fulfilled, and students learned useful skills in ESP32 programming, Bluetooth interface, sensor data collection, and Internet of Things-based control systems.

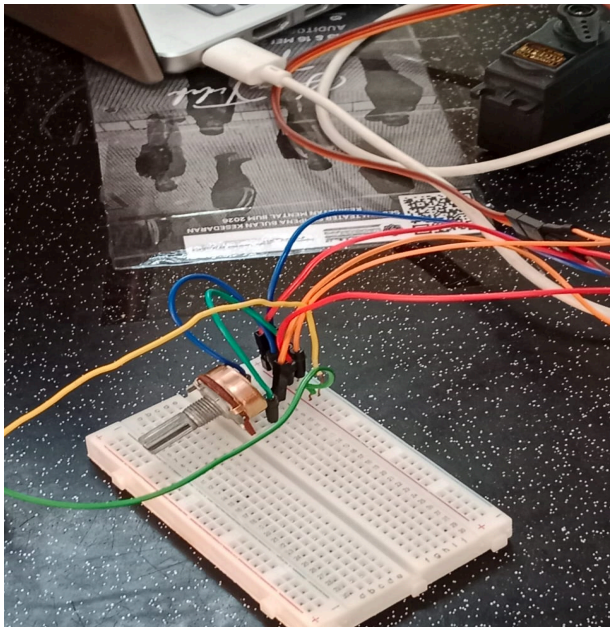
Recommendation

Several improvements can be implemented to improve the performance and functionality of the Bluetooth-based monitoring and control system. First, a more stable external power supply can be used for the servo motor because the ESP32 alone may not provide enough current during continuous servo operation. This improvement can help reduce unstable servo movement and prevent sudden system resets.

Second, the system can be enhanced by developing a mobile application with a graphical user interface (GUI) instead of relying only on the Serial Bluetooth Terminal application. A custom mobile application would make the system more attractive, user-friendly, and easier for users to control and monitor. In addition, extra sensors such as temperature, ultrasonic, or motion sensors can be integrated into the system to expand its Internet of Things (IoT) capabilities. With these additional sensors, the ESP32 can perform more advanced monitoring and automation tasks in different real-life applications.

Lastly, the response speed and accuracy of the system can be improved by optimizing the program code and minimizing unnecessary delay times in the loop function. This can provide smoother servo movement, faster communication between the ESP32 and smartphone, and overall better system performance.

Appendices



Appendix 1 : Circuit setup using servo motor and potentiometer as sensor

Code snippet :

<u>Arduino coding</u>

```
#include "BluetoothSerial.h"
#include <ESP32Servo.h> // Required for Task 1

BluetoothSerial SerialBT;
Servo myServo;

const int potpin = 34;    // Potentiometer pin
     // Onboard LED pin
const int servoPin = 13; // Suggested pin for Servo signal

char command;
int potValue = 0;

void setup()
{
  Serial.begin(9600);
  SerialBT.begin("Group_10"); // Replace with your group name
```

```
  myServo.attach(servoPin); // Initialize the servo

  Serial.println("Bluetooth initialized... Ready to receive commands.");
}


void loop()
{
  // Read potentiometer
  potValue = analogRead(potpin);

  // Convert pot value to angle
  int potAngle = map(potValue, 0, 4095, 0, 180);

  // Default mode: potentiometer control
  int angle = potAngle;

  // Check if Bluetooth command exists
  if (SerialBT.available())
  {
    command = SerialBT.read();
    Serial.print("Command received: ");
    Serial.println(command);

    switch (command)
    {
      case 'L': angle = 0;   break;   // Left
      case 'M': angle = 90;  break;   // Middle
      case 'R': angle = 180; break;   // Right
    }
  }

  // Move servo
  myServo.write(angle);

  // Send data to phone
  SerialBT.print("Pot: ");
  SerialBT.print(potValue);
```

```
  SerialBT.print(" | Angle: ");
  SerialBT.println(angle);

  delay(1000);
}
```

References

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First of all, we would like to sincerely thank our lecturer, Dr.-Ing. Wahju Sediono, for his guidance and support throughout this experiment. His explanations and advice really helped us understand how the ESP32 and a smartphone communicate using bluetooth, and IoT (Internet of Things) applications more clearly.

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Besides that, we are grateful to our classmates for their help and discussions, which made the learning process more enjoyable and meaningful.

Student's Declaration

Certificate of Originality and Authenticity

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgement, and that the original work contained herein have not been untaken or done by unspecified sources or persons.

We hereby certify that this report has **not been done by only one individual** and **all of us have contributed to the report.** The length of contribution to the reports by each individual is noted within this certificate.

We also hereby certify that we have **read** and **understand** the content of the total report and no further improvement on the reports is needed from any of the individual's contributors to the report.

We therefore, agreed unanimously that this report shall be submitted for **marking** and this **final printed report** has been **verified by us.**

Signature: Read [/]
Name: Nur Irdina Binti Omar Understand [/]
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Contribution: Abstract , Methodology, Recommendation

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Contribution: Introduction, Material equipment, Experimental setup, Conclusion

Signature: Read [/]
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Matric Number: 2416554 Agree [/]
Contribution: Data analysis, Data collection, Results, Discussion, References